

The DormTracker

II. Problem Description: When living in a dorm room with others, it can be difficult to keep track of chore assignments, room group supply fund debts, and neat little facts about each roommate. If room chores are not tracked, certain roommates will do more than others. If nothing is written down, shared debts will be lost in the shuffle. DormTracker allows users to create a room profile which automatically rotates chores and saves personal descriptions about each roommate in a digital almanac.

III. Goals:

Create Room & User Profile: When DormTracker first runs it will ask for the users dorm room number, their name, and each of their roommates names and a quick description.

Chore Rotator: Assign each person a cleaning chore everyday with no repetitions.

Almanac Look-up: Access any of your roommate's profile and description in less than 2 seconds.

IV. Features:

Startup Questions: When Dorm Tracker first opens, ask users for their room number, their name, how many roommates they have, each of their roommates names, and a quick description to save about each of them.

Roommate Almanac: Allows users to look up any of their roommate's profile information and description that they saved.

Automatic Chore Scheduler: Assign each roommate a different cleaning chore in an equal rotation.

Supply Debt Tracker: Keep track of any collective supply purchases the room makes and calculate who is owes whom and how much.

V. Inputs and Outputs:

Inputs: Room number, user name, number of roommates, roommate names, description of roommates, list of cleaning chores, cost of bought item, name of person who paid for item.

Outputs: Welcome message with room number, formatted view of roommate's profile and description, board member with a cleaning chore, message stating who owes whom and how much, main menu option.

VI. Pseudocode:

START

PROMPT user for room_number

PROMPT user for user_name

PROMPT user for total_roommates

FOR i **FROM** 1 **TO** total_roommates **DO**

PROMPT for roommate_name

PROMPT for roommate_description

STORE roommate_name and roommate_description in almanac_dictionary

ENDFOR

SET running = **TRUE**

WHILE running **IS TRUE DO**

DISPLAY "=== DORMTRACKER: ROOM " + room_number + " ==="

DISPLAY "User: " + user_name

DISPLAY "[1] Roommate Almanac"

DISPLAY "[2] Manage Cleaning Chores"

DISPLAY "[3] Log Shared Expense"

DISPLAY "[4] View Summary & Balances"

DISPLAY "[5] Exit Program"

PROMPT user for choice

IF choice == 1 **THEN**

DISPLAY list of registered roommates

PROMPT user to select a roommate name

IF selected name **EXISTS** in almanac_dictionary **THEN**

DISPLAY roommate_name + ": " + stored roommate_description

ELSE

DISPLAY "Error: Roommate not found."

ENDIF

ELSE IF choice == 2 **THEN**

PROMPT user for list of chores

FOR each chore **DO**

ASSIGN chore to roommate using round-robin index

ENDFOR

DISPLAY generated cleaning schedule

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ELSE IF choice == 3 THEN
    GET expense item name, total cost, AND payer name
    IF cost <= 0 THEN
        DISPLAY "Error: Cost must be a positive number."
    ELSE
        CALCULATE split_amount = total cost / total_roommates
        UPDATE balances for each roommate
        DISPLAY "Expense logged successfully."
    ENDIF

ELSE IF choice == 4 THEN
    DISPLAY overall room summary, chore schedule, and balances

ELSE IF choice == 5 THEN
    DISPLAY "Exiting DormTracker. Goodbye!"
    SET running = FALSE

ELSE
    DISPLAY "Invalid choice. Please select 1 to 5."
ENDIF

ENDWHILE
END
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